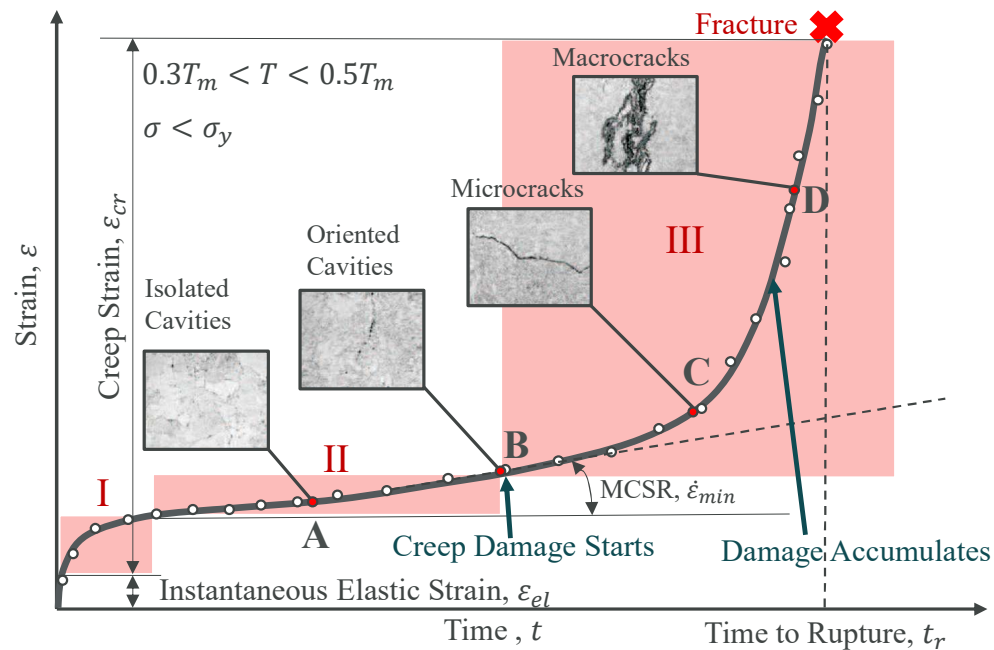


# Creep and Creep Induced Failure



## I - Primary Creep:

**Hardening Processes**, obstruction of the dislocation movements,  
**Relaxation Processes**, e.g. redistribution of lattice defects

## II - Secondary Creep:

**Hardening & Softening Processes**, are in equilibrium,  
**Steady-state**, or minimum-creep-strain-rate,  $\dot{\epsilon}_{min}$

## III - Tertiary Creep:

**Damage Processes**, e.g. formation, growth & coalescence of voids at the grain boundaries, Aging of the microstructure, etc.

A – Observation

B – Observation, fixed inspection intervals

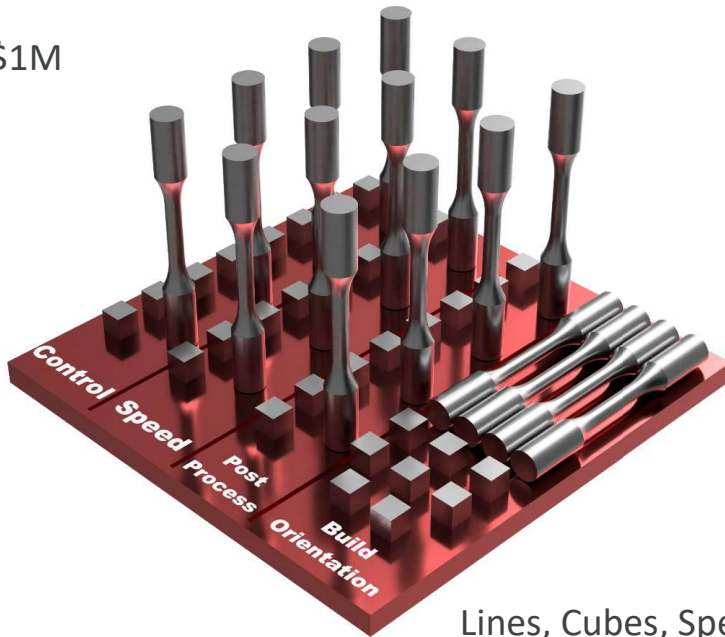
C – Limited service until repair

D – Immediate repair

# The Need...

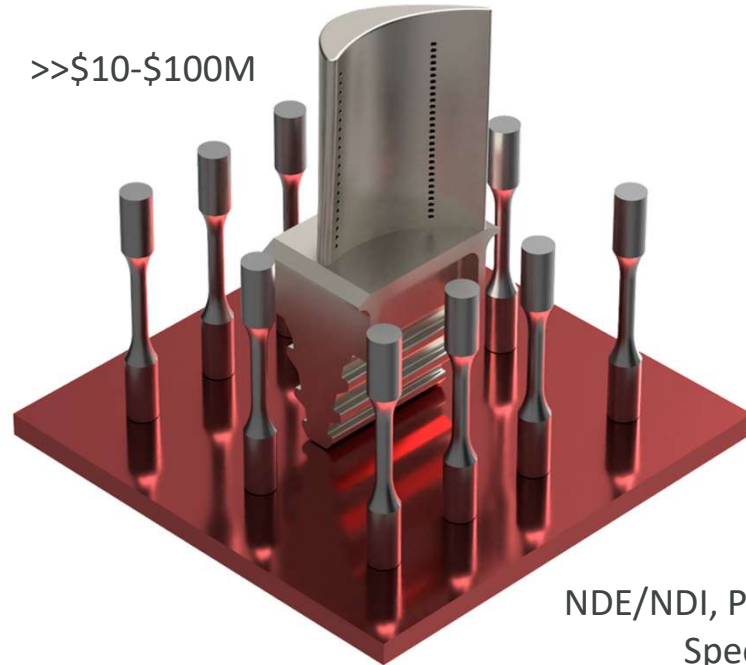
## Operational Qualification (OQ)

≈\$1M



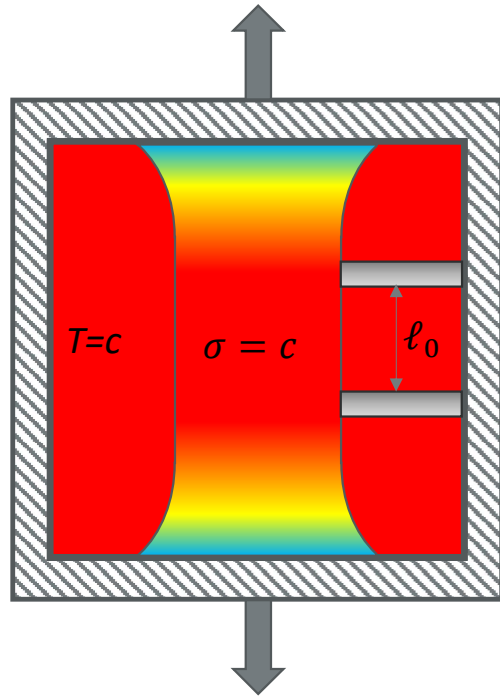
## Performance Qualification (PQ)

>>\$10-\$100M



How can we lower the **costs** and shorten the **timeline** for new materials discovery (MGI) and AM qualification (IQ/OQ/PQ)?

# ASTM Standard Testing



Numerous tests are required to measure the high-temperature **material properties** over a component's service conditions ( $\Delta\sigma, \Delta T$ ).

1 test = 1 data point

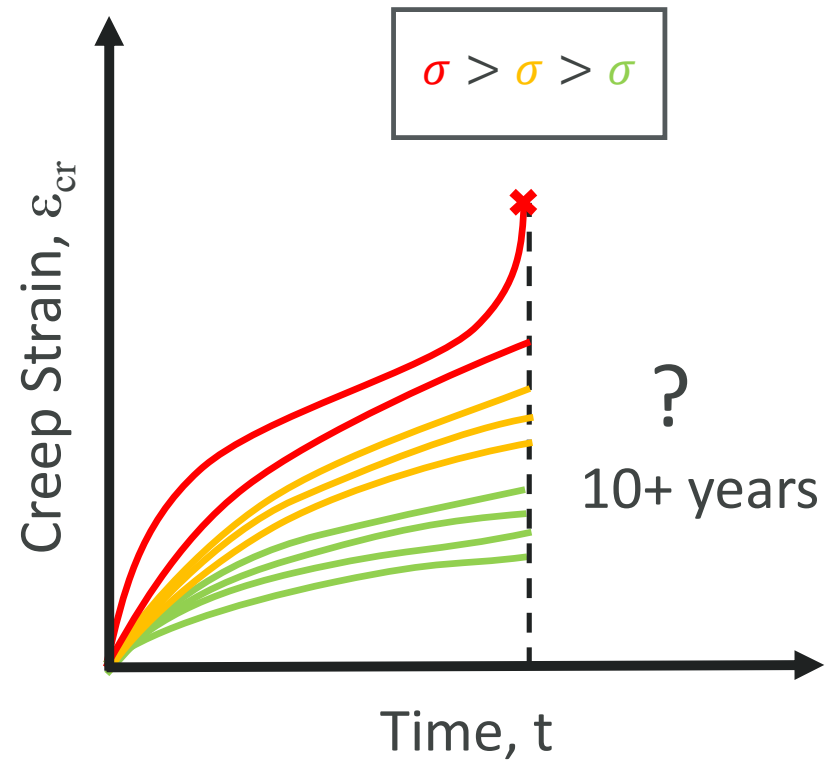
There are stringent requirements

- Uniform Temperature,  $T = \text{const.}$
- Uniform Stress,  $\sigma = \text{const.}$
- Load cell accuracy
- Extensometer classification (A, B1, B2, C)
- Replicate/Repeatability Testing

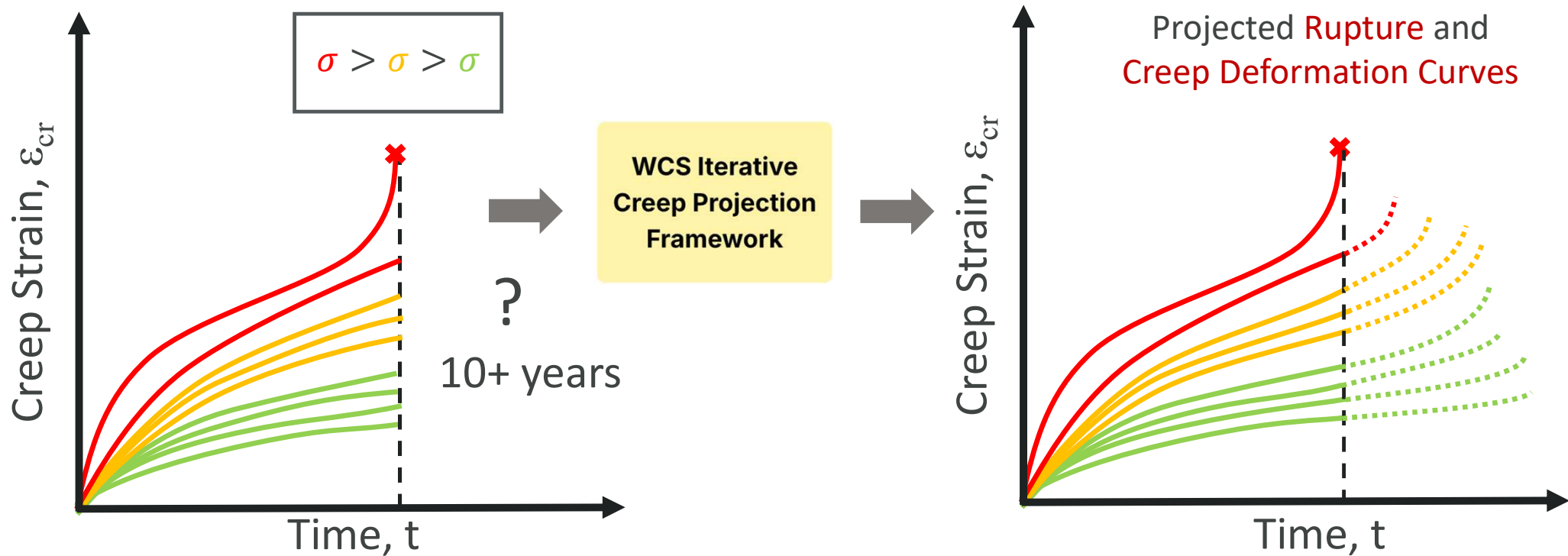
# Managing Long-Duration, High-Uncertainty Creep Tests



The National Research Institute for Metals in Japan



# A Data-Driven Creep Forecasting Tool



# Constitutive Models

- **McVetty Primary Creep** is a time hardening model and captures the initial, decelerating stage of creep deformation

Primary Creep Strain

$$\varepsilon_{pr} = G \cdot (1 - e^{-q \cdot t})$$

- **Wilshire-Cano-Stewart (WCS)** is a continuum damage mechanics based constitutive model consist of coupled creep-strain-rate and damage-rate equations

Creep-strain-rate

$$\dot{\varepsilon}_{cr} = \frac{\left[ -\frac{\ln(\sigma/\sigma_{TS})}{k_2} \right]^{\frac{1}{v}}}{\exp(Q_c/RT)} \exp(\omega)$$

Damage-rate equation

$$\dot{\omega} = \frac{[1 - \exp(-\varphi)]}{\varphi} \frac{\exp(-Q_c/RT)}{\left[ -\frac{\ln(\sigma/\sigma_{TS})}{k_1} \right]^{\frac{1}{u}}} \exp(\lambda\omega)$$

| Material Constant | Behavior                         |
|-------------------|----------------------------------|
| $R$               | Molar Gas Constant               |
| $T$               | Absolute Temperature             |
| $\sigma_{TS}$     | Ultimate Tensile Strength        |
| $Q_c$             | Apparent Activation Energy       |
| $k_1, u, k_2, v$  | Wilshire Model Constants         |
| $\phi$            | Damage Trajectory Constant       |
| $\lambda$         | Creep Strain Trajectory Constant |
| $\omega$          | Internal State Variable, Damage  |
| $G$               | Max Primary Creep Strain         |
| $q$               | Primary Creep Exhaustion Rate    |

# Wilshire-Cano-Stewart (WCS) Model

## Rate Based

$$\dot{\varepsilon}_{cr} = \frac{\left[ -\frac{\ln(\sigma/\sigma_{TS})}{k_2} \right]^{\frac{1}{v}}}{\exp(Q_c/RT)} \exp(\omega)$$



## Closed Form

$$\varepsilon(t) = \dot{\varepsilon}_{min} \left( \frac{\phi \left( (t - t_r)e^\phi - t \right) \left( \frac{t(e^{-\phi} - 1)}{t_r} + 1 \right)^{\frac{-\lambda}{\phi}} + t_r \phi e^\phi}{(\phi - \lambda)(e^\phi - 1)} \right) + \varepsilon_{pr}$$

Secondary + Tertiary Creep

Primary Creep

$$\dot{\omega} = \frac{[1 - \exp(-\phi)] \exp(-Q_c/RT)}{\phi} \frac{\exp(\lambda\omega)}{\left[ -\frac{\ln(\sigma/\sigma_{TS})}{k_1} \right]^{\frac{1}{u}}}$$

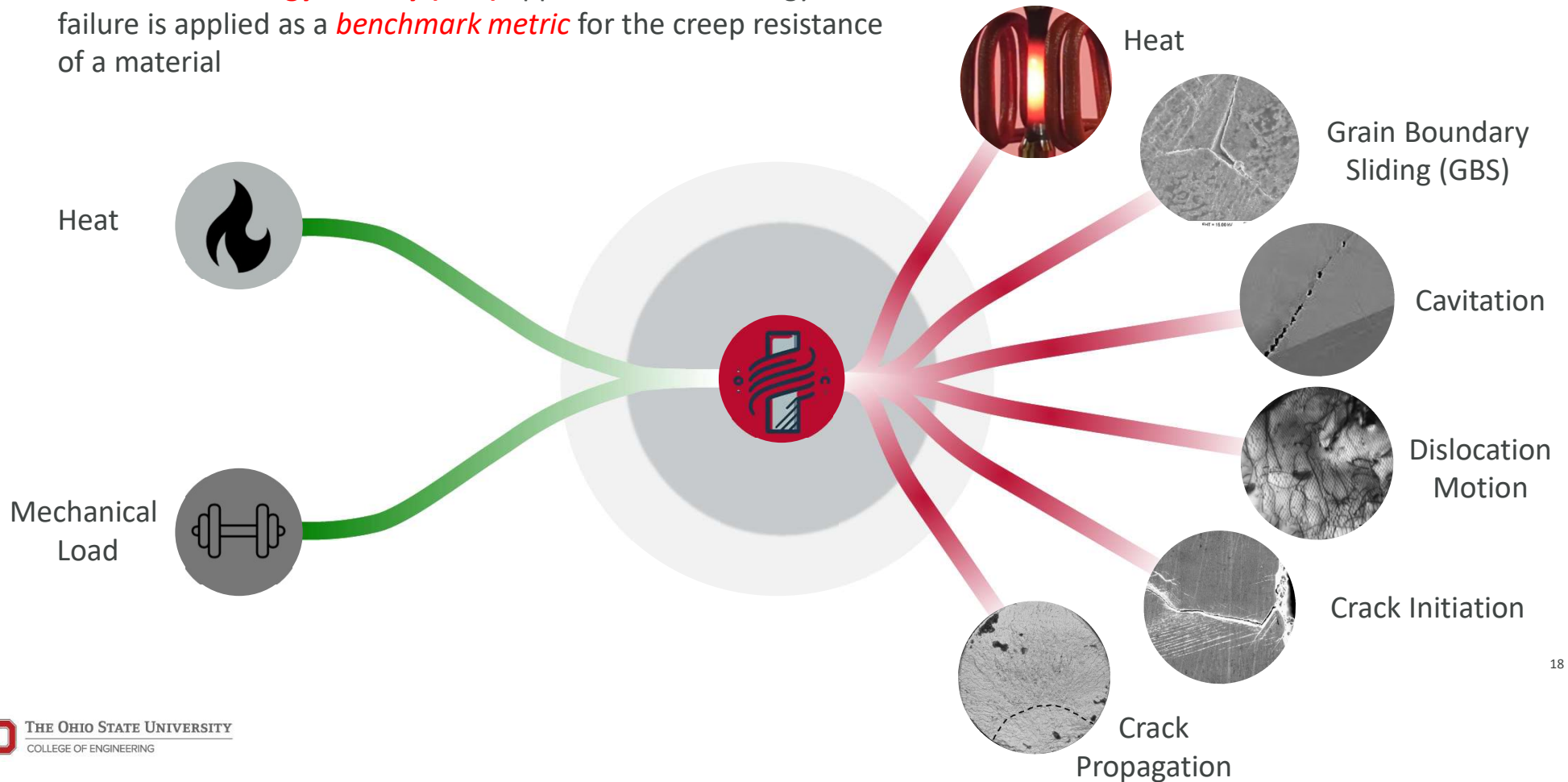


$$\omega(t) = \frac{1}{-\phi} \ln \left[ 1 - [\exp(-\phi)] \frac{t}{t_r} \right]$$



# Strain Energy Density Dissipation (SED)

In the **Strain Energy Density (SED)** approach, strain energy-to-failure is applied as a *benchmark metric* for the creep resistance of a material





# Strain Energy Density Dissipation (SED)

In the **Strain Energy Density (SED)** approach, assume that a material has a finite amount of energy it can dissipate through plastic deformation until reaching rupture

$$W_f = \sigma \varepsilon_f = \int_0^{t_r} \sigma(t) \dot{\varepsilon} dt$$

SED expressed as cumulative experimental damage

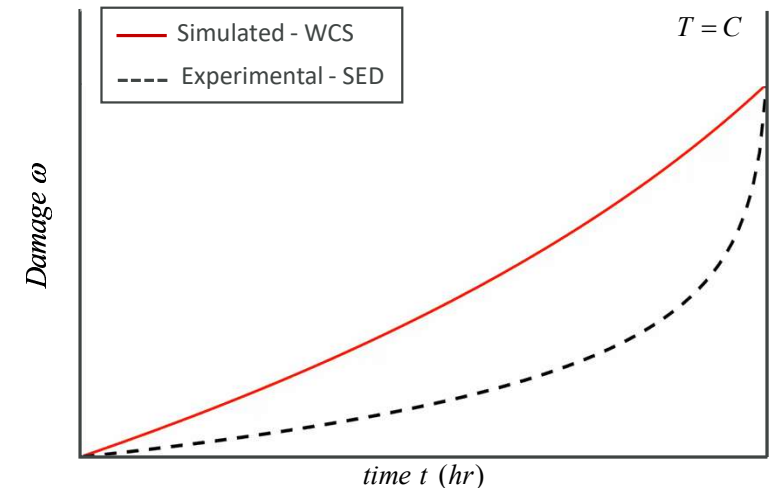
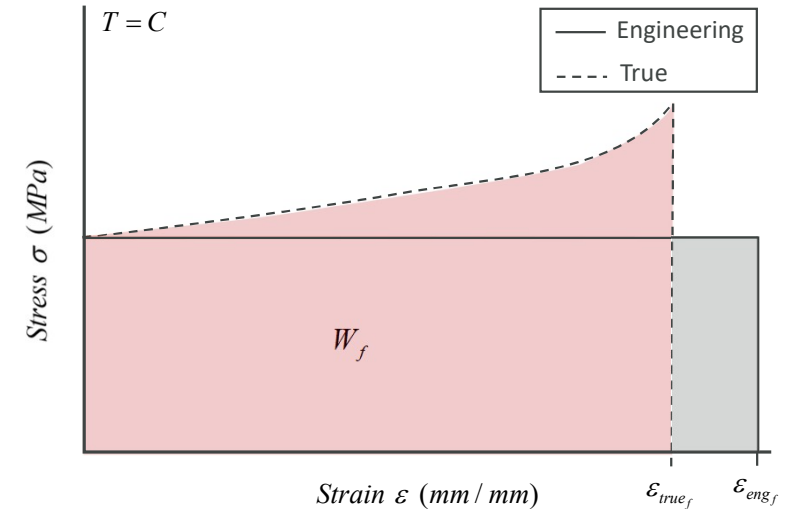
$$\omega = \frac{W}{W_f}, \quad 0 \leq \omega \leq 1 \quad \text{where } 0 \text{ (undamaged) and } 1 \text{ (rupture)}$$

A single  $\phi$  is determined to satisfy the damage evolution described using SED

$$\omega(t) = -\frac{1}{\phi} \ln \left[ 1 - [1 - \exp(-\phi)] \frac{t}{t_r} \right] \quad \text{where } \phi \geq 0$$

## Advantages

- Removes stress and temperature sensitivity
- Provides an experimentally realistic damage to improve modeling repeatability and accuracy



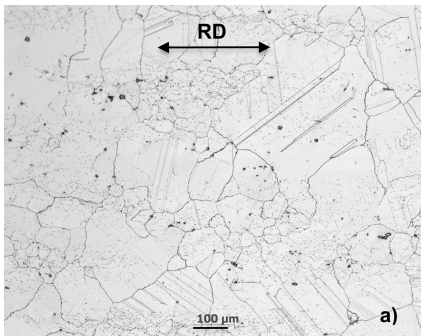
# Material: Alloy 617

Alloy 617 is Ni-Cr-Mo-Co based material

Strengthened by solid solution hardening from:

- Chromium
- Molybdenum
- Cobalt
- Intra- and inter-granular carbide precipitates

**Applications:** Gas turbines, high-temperature nuclear reactors

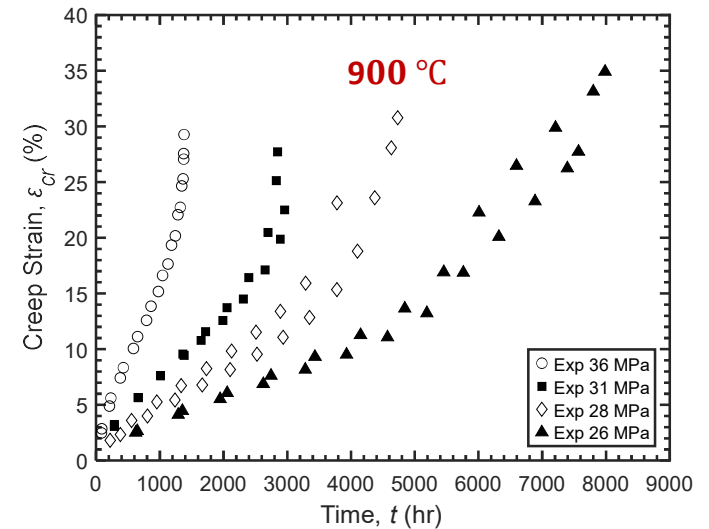
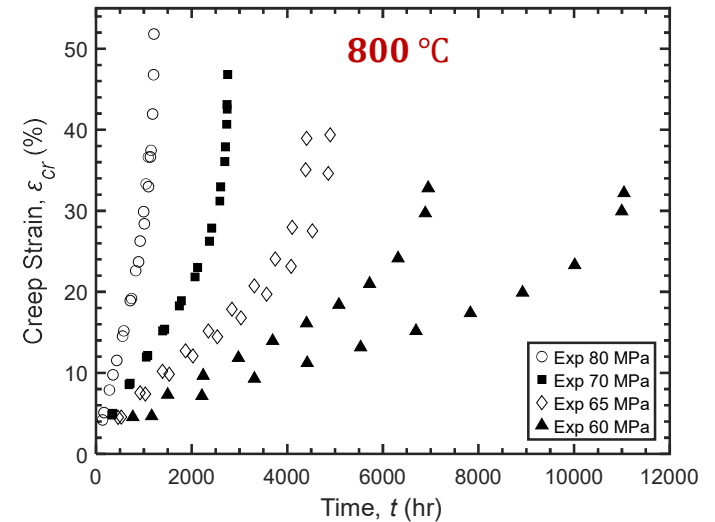


Alloy 617 Chemical Composition

| Element | Wt. %  | Nominal Wt. % |
|---------|--------|---------------|
| Ni      | 54.1   | 44.5 min      |
| Cr      | 22.2   | 20-24         |
| Co      | 11.6   | 10-15         |
| Mo      | 8.6    | 8-10          |
| Fe      | 1.6    | 3 max         |
| Mn      | 0.1    | 1 max         |
| Al      | 1.1    | 0.8-1.5       |
| C       | 0.05   | 0.05-0.15     |
| Cu      | 0.04   | 0.5 max       |
| Si      | 0.1    | 1 max         |
| S       | <0.002 | 0.015 max     |
| Ti      | 0.4    | 0.6 max       |
| B       | <0.001 | 0.006 max     |

J. Wright, Draft ASME Boiler and Pressure Vessel Code Case for Use of Alloy 617 for Class A Elevated Temperature Service Construction, INL/EXT-15-36305, Idaho National Laboratory, 2015.

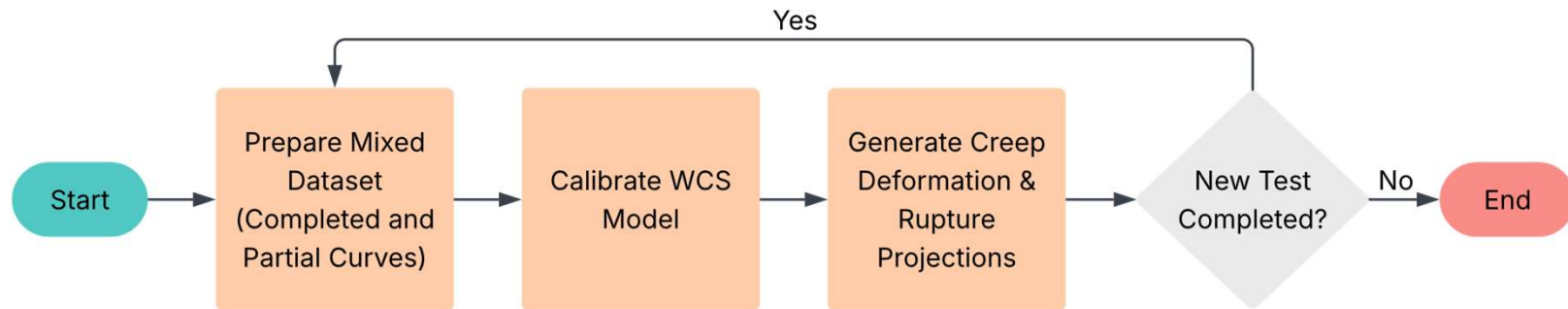
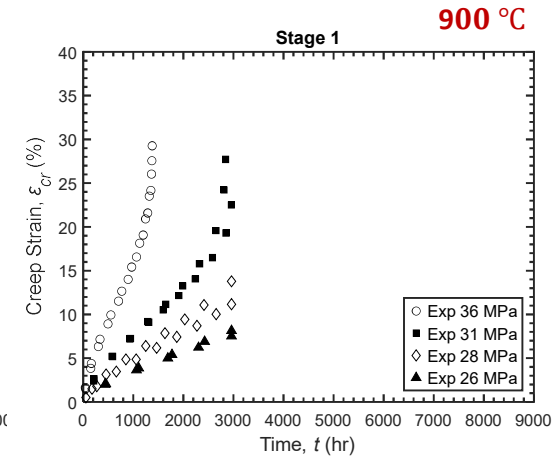
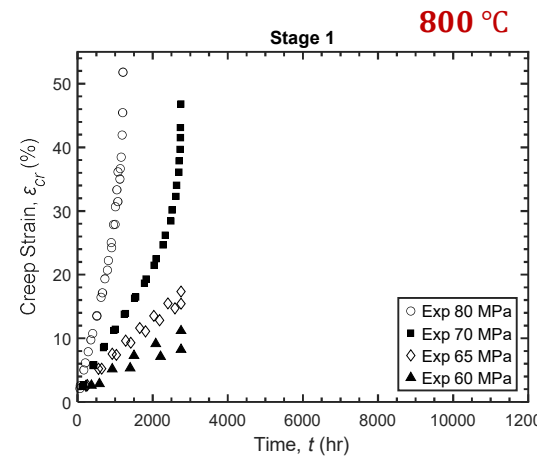
## Creep Deformation



# Experiment Test Matrix

Stage 1

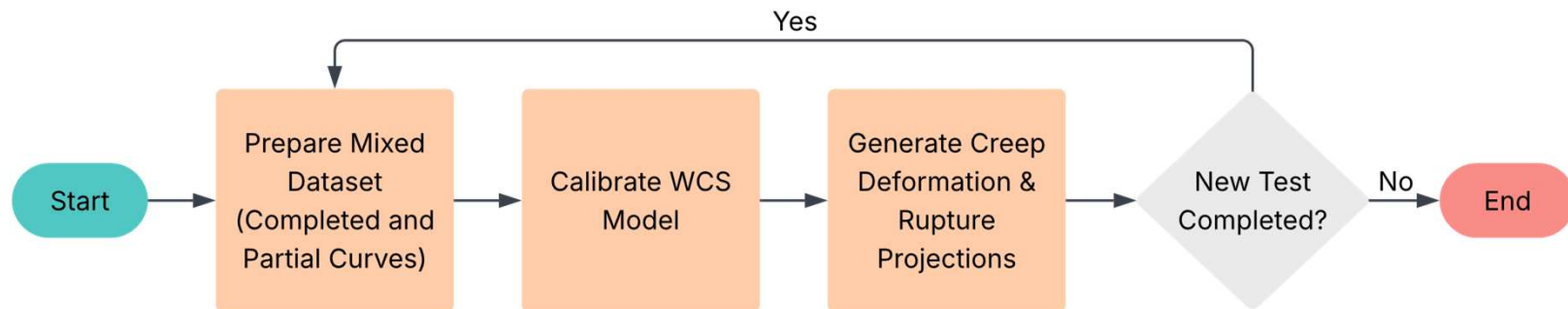
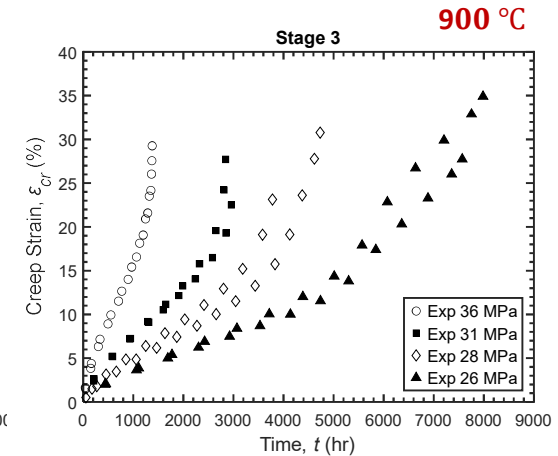
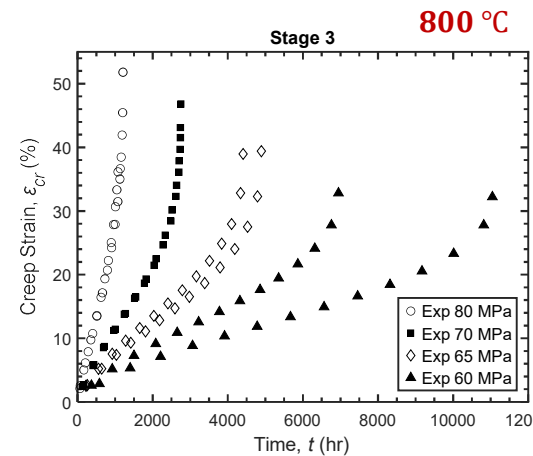
|                  |              | WCS Calibration Stage |          |          |
|------------------|--------------|-----------------------|----------|----------|
| Temperature (°C) | Stress (MPa) | 1                     | 2        | 3        |
| 800              | 80           | Complete              | Complete | Complete |
|                  | 70           | Complete              | Complete | Complete |
|                  | 65           | Partial               | Complete | Complete |
|                  | 60           | Partial               | Partial  | Complete |
| 900              | 36           | Complete              | Complete | Complete |
|                  | 31           | Complete              | Complete | Complete |
|                  | 28           | Partial               | Complete | Complete |
|                  | 26           | Partial               | Partial  | Complete |



# Experiment Test Matrix

Stage 3

|                  |              | WCS Calibration Stage |          |          |
|------------------|--------------|-----------------------|----------|----------|
| Temperature (°C) | Stress (MPa) | 1                     | 2        | 3        |
| 800              | 80           | Complete              | Complete | Complete |
|                  | 70           |                       |          |          |
|                  | 65           | Partial               |          |          |
|                  | 60           |                       |          |          |
| 900              | 36           | Complete              | Complete | Complete |
|                  | 31           |                       |          |          |
|                  | 28           | Partial               |          |          |
|                  | 26           |                       |          |          |

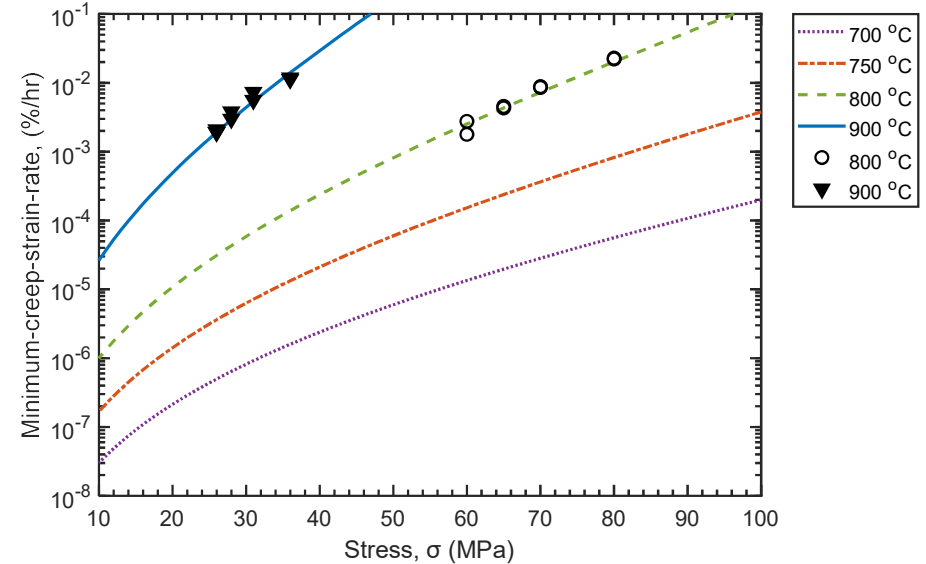
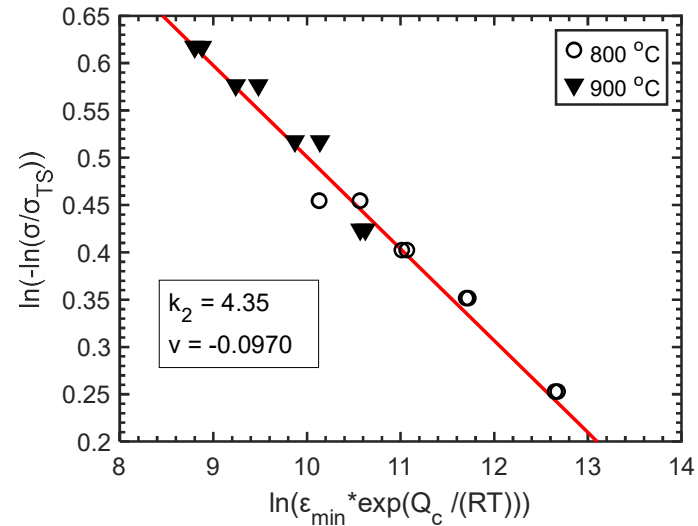


# Wilshire Constants

Minimum-creep-strain-rate data is used to numerically calibrate Wilshire constants  $k_2$ ,  $v$  and Activation Energy  $Q_c$

$$\dot{\epsilon}_{min} = \frac{\left[ -\frac{\ln\left(\frac{\sigma}{\sigma_{TS}}\right)}{k_2} \right]^{\frac{1}{v}}}{\exp\left(\frac{Q_c}{RT}\right)}$$

| Temperature (°C) | Stress (MPa) | $\dot{\epsilon}_{min}$ (%hr <sup>-1</sup> ) | $k_2$ (hr <sup><math>v</math></sup> ) | $v$ (unitless) | $Q_c$ kJ · mol <sup>-1</sup> |
|------------------|--------------|---------------------------------------------|---------------------------------------|----------------|------------------------------|
| 800              | 80           | $2.01 \times 10^{-2}$                       | 4.35                                  | -0.097         | 147                          |
|                  | 70           | $7.28 \times 10^{-3}$                       |                                       |                |                              |
|                  | 65           | $4.31 \times 10^{-3}$                       |                                       |                |                              |
|                  | 60           | $2.52 \times 10^{-3}$                       |                                       |                |                              |
| 900              | 36           | $1.40 \times 10^{-2}$                       |                                       |                |                              |
|                  | 31           | $5.35 \times 10^{-3}$                       |                                       |                |                              |
|                  | 28           | $2.91 \times 10^{-3}$                       |                                       |                |                              |
|                  | 26           | $1.91 \times 10^{-3}$                       |                                       |                |                              |



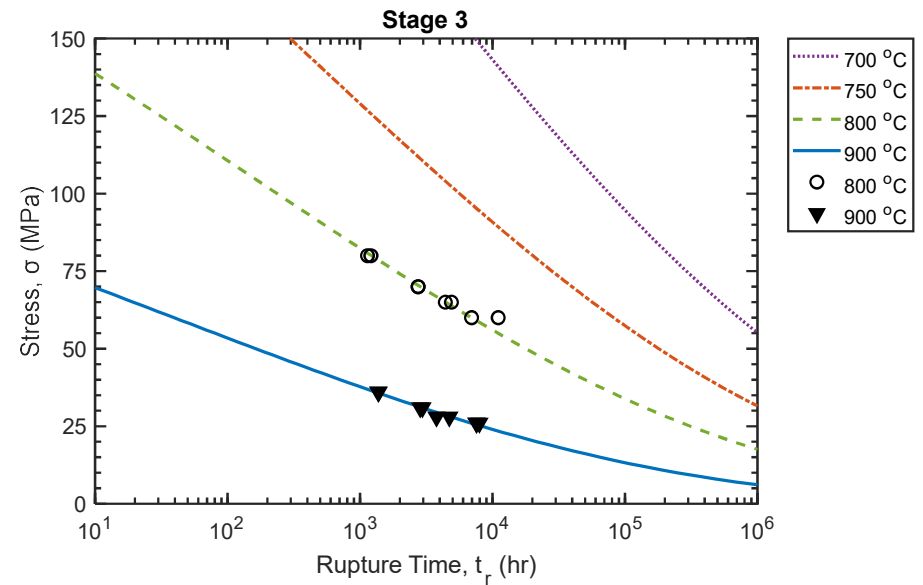
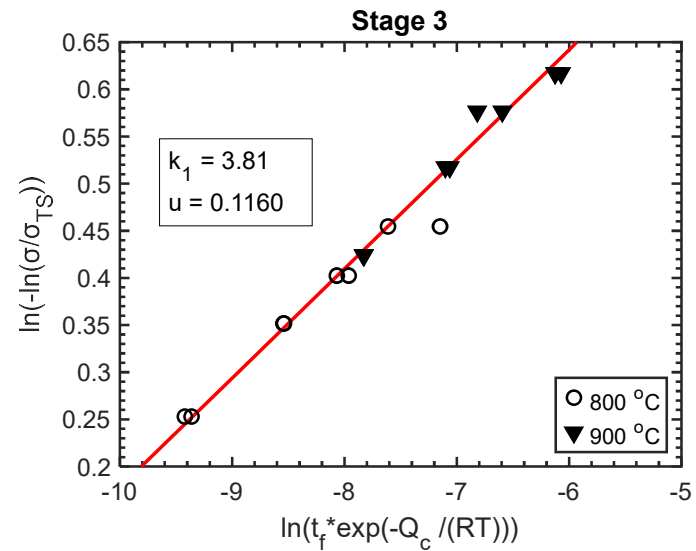
# Wilshire Constants

Creep stress rupture data and Activation Energy is used to numerically calibrate Wilshire constants  $k_1$  and  $u$

$$t_r = \frac{\left[ -\frac{\ln\left(\frac{\sigma}{\sigma_{TS}}\right)}{k_1} \right]^{\frac{1}{u}}}{\exp\left(\frac{-Q_C}{RT}\right)}$$

| Temperature (°C) | Stress (MPa) | $\overline{Exp. t_r}$ (hr) | $Pred. t_r$ (hr) | $k_1$ (hr <sup>-u</sup> ) | $u$ (unitless) |
|------------------|--------------|----------------------------|------------------|---------------------------|----------------|
| 800              | 80           | 1173                       | 1220             | 3.81                      | 0.1160         |
|                  | 70           | 2745                       | 2856             |                           |                |
|                  | 65           | 4650                       | 4425             |                           |                |
|                  | 60           | 8998                       | 6935             |                           |                |
| 900              | 36           | 1377                       | 1313             |                           |                |
|                  | 31           | 2903                       | 2934             |                           |                |
|                  | 28           | 4256                       | 4874             |                           |                |
|                  | 26           | 7776                       | 6928             |                           |                |

Stage 3

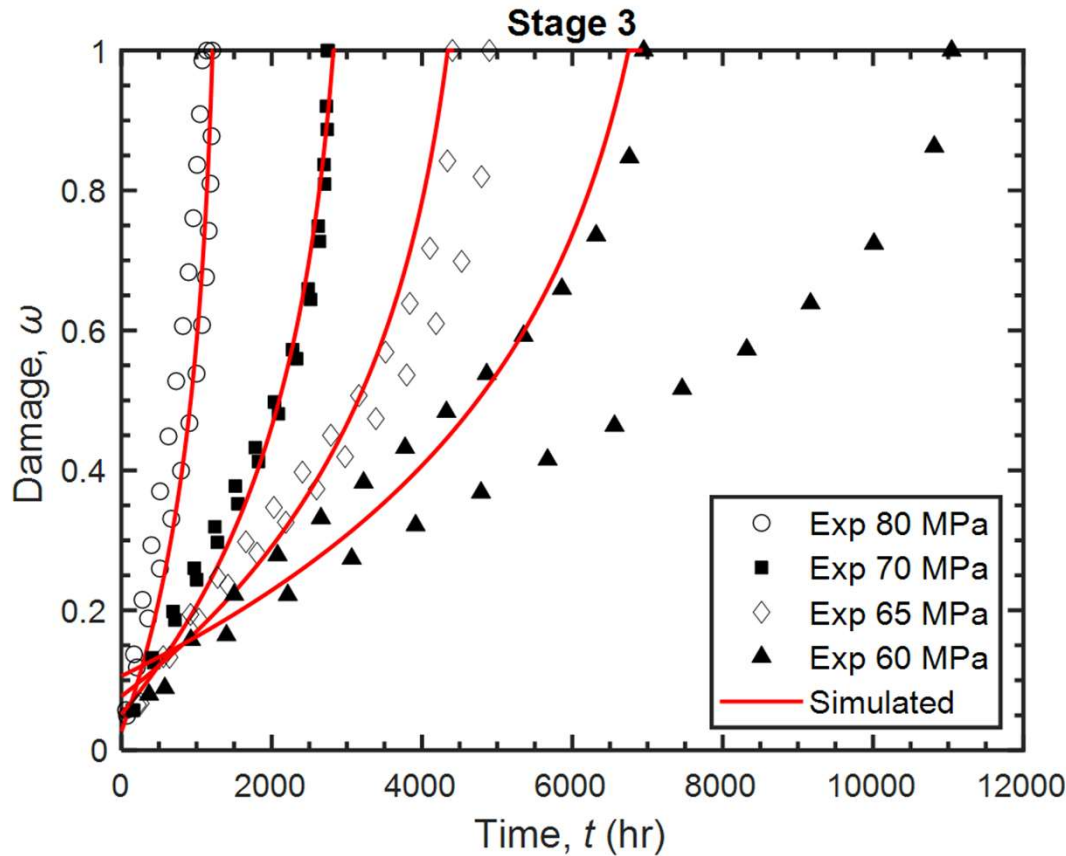


# Damage Predictions at 800°C

Stage 1

Stage 2

Stage 3



Experimental Damage

$$\omega = \frac{w}{w_f}, 0 \leq \omega \leq 1$$

Modeled Damage

$$\omega(t) = \frac{1}{-\phi} \ln \left[ 1 - [\exp(-\phi)] \frac{t}{t_r} \right]$$

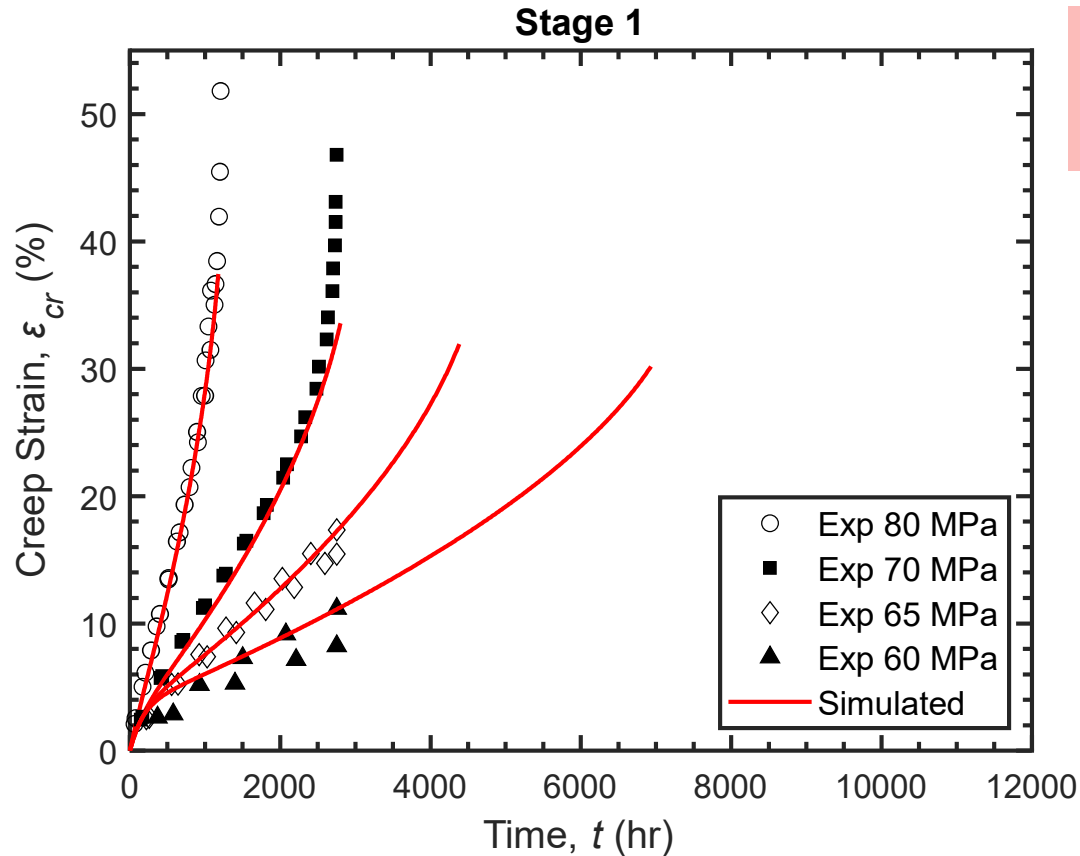
Root Mean Squared Error Minimized

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (model_i - exp_i)^2}$$

|                      |              | WCS Calibration Stage |          |          |
|----------------------|--------------|-----------------------|----------|----------|
| Temperature (°C)     | Stress (MPa) | 1                     | 2        | 3        |
| 800                  | 80           | Complete              | Complete | Complete |
|                      | 70           |                       |          |          |
|                      | 65           | Partial               | Partial  |          |
|                      | 60           |                       |          |          |
| 900                  | 36           | Complete              | Complete | Complete |
|                      | 31           |                       |          |          |
|                      | 28           | Partial               | Partial  |          |
|                      | 26           |                       |          |          |
| Calibrated Phi (ϕ) : |              | 2.72                  | 2.77     | 2.51     |



# Creep Strain Predictions at 800°C



## Modeled Creep Strain

$$\varepsilon(t) = \dot{\varepsilon}_{min} \left( \frac{\phi \left( (t - t_r)e^\phi - t \right) \left( \frac{t(e^{-\phi} - 1)}{t_r} + 1 \right)^{\frac{-\lambda}{\phi}} + t_r \phi e^\phi}{(\phi - \lambda)(e^\phi - 1)} \right) + \varepsilon_{pr}$$

## Root Mean Squared Error Minimized

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (model_i - exp_i)^2}$$

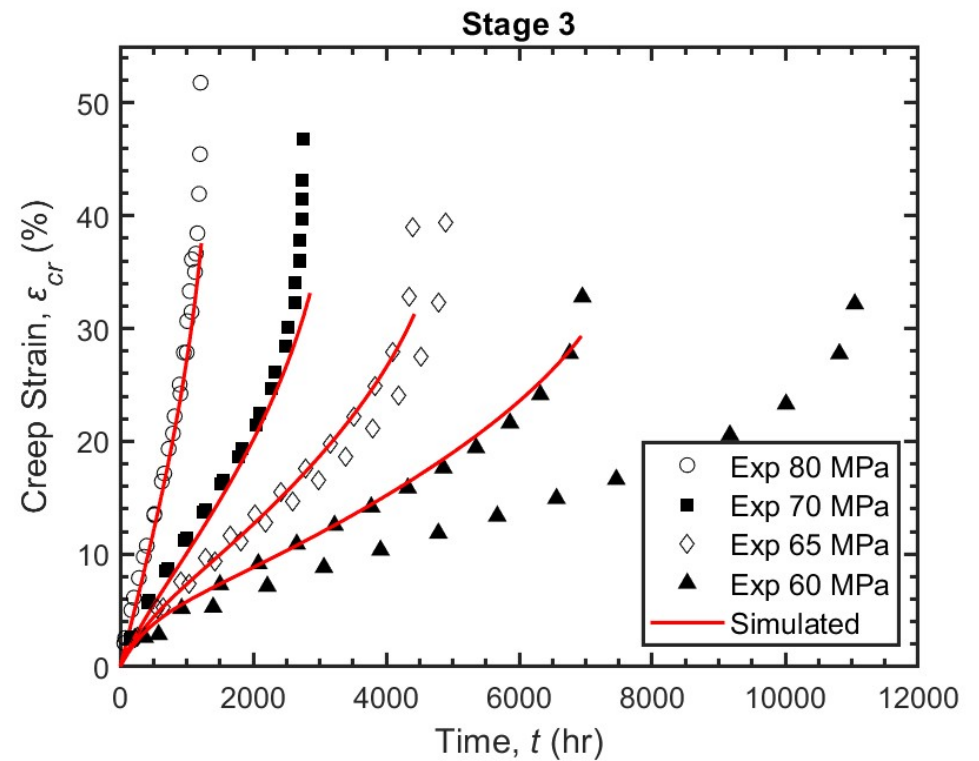
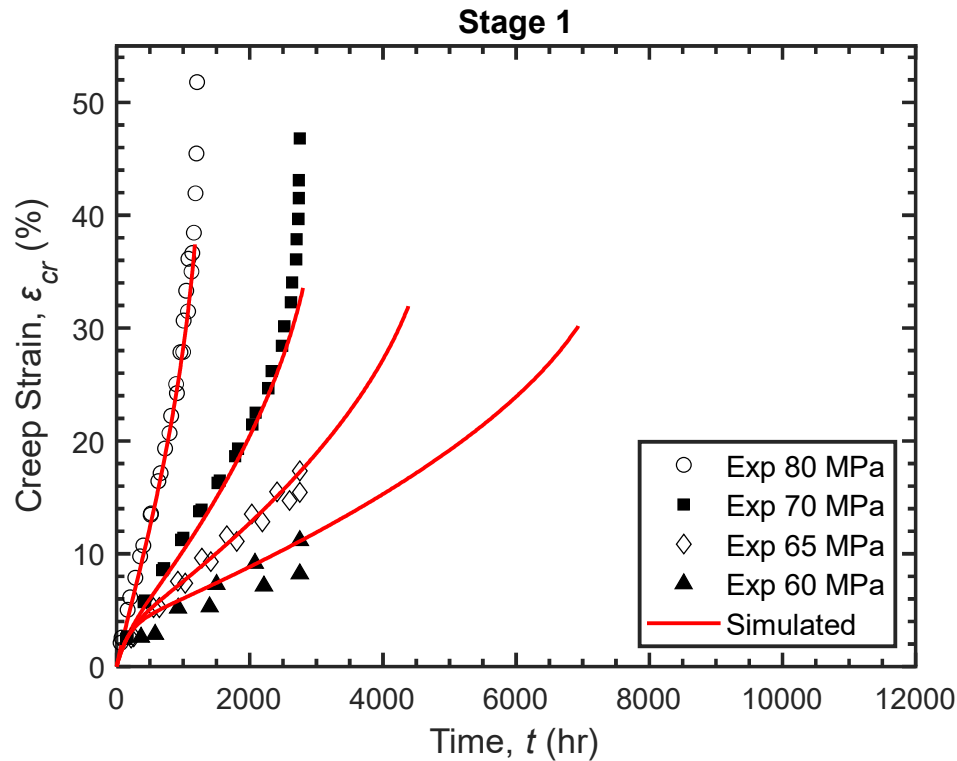
## WCS Calibration Stage

| Temperature (°C) | Stress (MPa) | 1        | 2        | 3        |
|------------------|--------------|----------|----------|----------|
| 800              | 80           | Complete | Complete | Complete |
|                  | 70           |          |          |          |
|                  | 65           | Partial  | Partial  |          |
|                  | 60           |          |          |          |
| 900              | 36           | Complete | Complete | Complete |
|                  | 31           |          |          |          |
|                  | 28           | Partial  | Partial  |          |
|                  | 26           |          |          |          |

Calibrated Lambda ( $\lambda$ ) :      **1.26**                      **1.29**                      **1.13**

# Creep Strain Predictions at 800°C

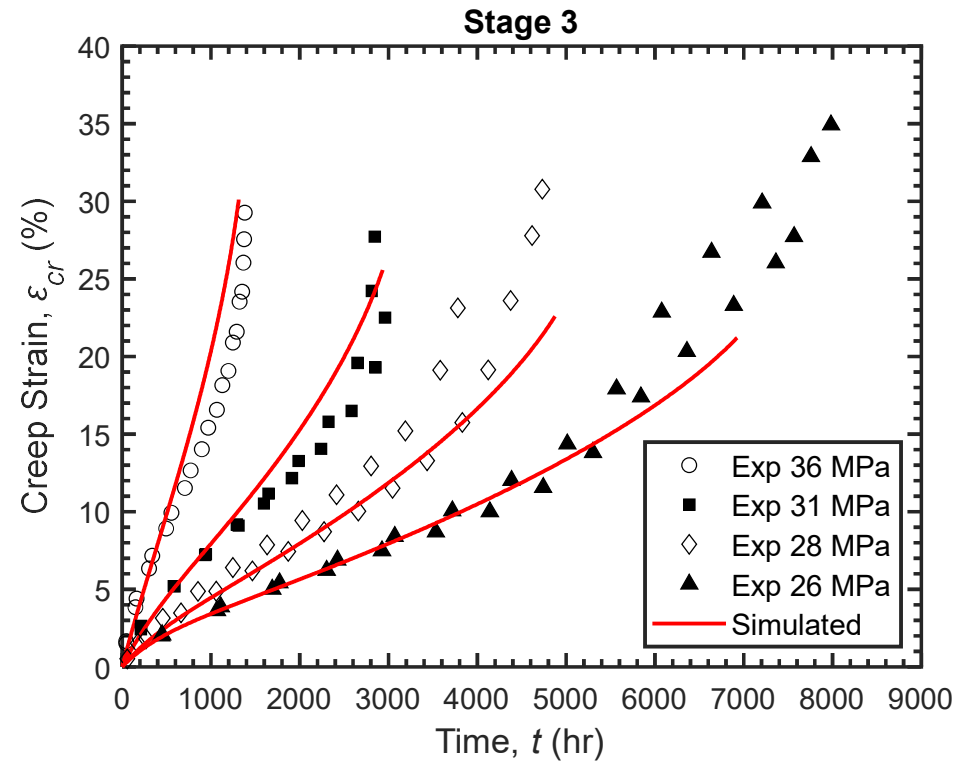
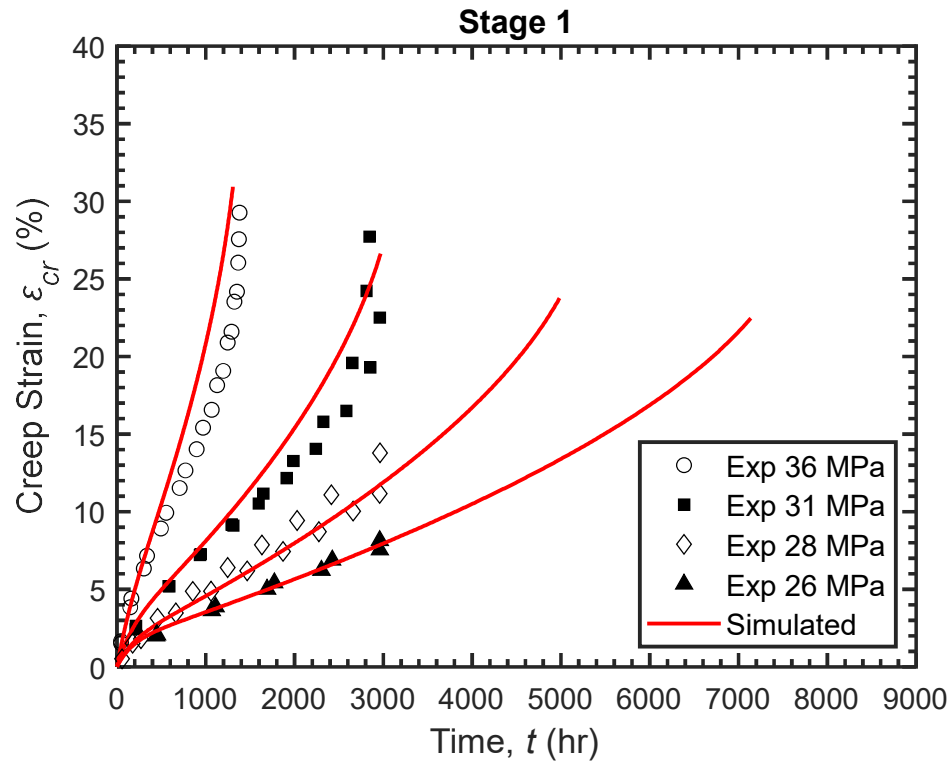
Stage 1 Stage 3



# Creep Strain Predictions at 900°C

Stage 1

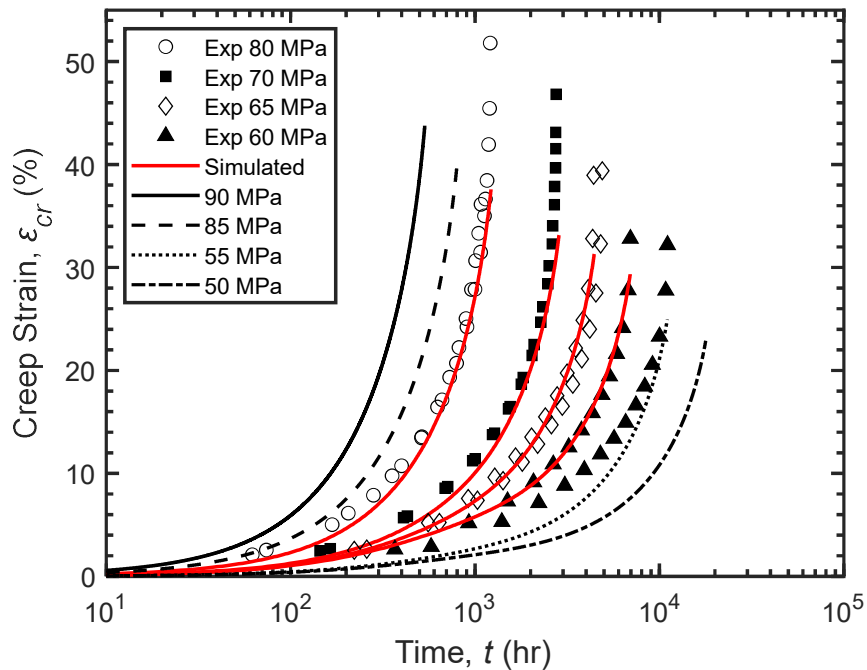
Stage 3



# Post Audit Validation

- WCS calibrated constants **isostress simulations** for post audit-validation
- Predictions are in good agreement with experimental data

**800°C**



**900°C**

